

# CPSC 250

## Craps Game Odds

Edward Brash

### 1 Motivation

The craps-style dice game used in the class example has an interesting property:

The probability of winning one credit is exactly equal to the probability of losing one credit.

That is:

$$P(+1) = P(-1) = \frac{1}{2}$$

This means the game behaves like a random walk.

### 2 Game Rules

On the initial roll:

- rolling 7 or 11 wins one credit;
- rolling 3 or 12 loses one credit;
- any other roll sets a goal and requires more rolling.

On later rolls:

- rolling 7 loses one credit;
- rolling the goal wins one credit;
- anything else means roll again.

### 3 Initial Roll Probabilities

The number of possible outcomes when rolling two dice is:

$$36$$

The probabilities of common sums are:

$$P(7) = \frac{6}{36} = \frac{1}{6}$$

$$P(11) = \frac{2}{36} = \frac{1}{18}$$

$$P(3) = \frac{2}{36} = \frac{1}{18}$$

$$P(12) = \frac{1}{36}$$

Therefore:

$$P(7 \text{ or } 11) = \frac{1}{6} + \frac{1}{18} = \frac{4}{18} = \frac{2}{9}$$

and:

$$P(3 \text{ or } 12) = \frac{1}{18} + \frac{1}{36} = \frac{3}{36} = \frac{1}{12}$$

## 4 Re-Roll Cases

If the initial roll is neither an immediate win nor an immediate loss, the game enters a re-roll state.

For the simplified version in the notes, this leads to a repeated process:

- some rolls end the round,
- some rolls continue the round.

The repeated continuation creates a geometric series.

## 5 Geometric Series Review

A geometric series has the form:

$$a + ar + ar^2 + ar^3 + \dots$$

If  $|r| < 1$ , then:

$$a + ar + ar^2 + ar^3 + \dots = \frac{a}{1 - r}$$

Equivalently:

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1 - r}$$

## 6 Probability of Eventually Winning

The notes derive the probability of winning as a sum involving immediate wins plus delayed wins.

The structure is:

$$P(+1) = \text{immediate win} + \text{delayed win}$$

The delayed portion includes the possibility of rolling again and again before eventually ending the round.

This repeated structure is exactly why a geometric series appears.

## 7 Final Result

After simplifying the geometric series, the result is:

$$P(+1) = \frac{1}{2}$$

Similarly:

$$P(-1) = \frac{1}{2}$$

Since every completed round either wins or loses one credit,

$$P(+1) + P(-1) = 1$$

and therefore, once one probability is known to be  $1/2$ , the other must also be  $1/2$ .

## 8 Interpretation

The game is fair in the sense that:

$$P(+1) = P(-1)$$

But this does not mean the game lasts forever.

The player's credits move up and down like a random walk until eventually reaching zero.

## 9 Random Walk Model

If the player begins with some number of credits, then after each round:

$$\text{credits} \rightarrow \text{credits} + 1$$

with probability  $1/2$ , or:

$$\text{credits} \rightarrow \text{credits} - 1$$

with probability  $1/2$ .

The game ends when:

$$\text{credits} = 0$$

## 10 Simulation Philosophy

The notes then transition from probability to simulation.

Instead of only solving analytically, we can simulate the game many times.

The idea is:

1. Run the game to completion many times.
2. Record the number of rounds in each game.
3. Record credits as a function of round number.
4. Use those results for analysis and plotting.

## 11 Data Structures for Simulation

For many simulations, we might use lists such as:

```
credit_result = []
round_result = []
result_number = []
```

For each simulated game, we append:

- the list of credit values,
- the list of round numbers,
- the final number of rounds.

## 12 Simulation Loop

A simplified structure is:

```
num_sims = 1000

credit_result = []
round_result = []
result_number = []

for i in range(num_sims):

    credit_round = []
    result_round = []

    # run one complete game here

    round_result.append(result_round)
    credit_result.append(credit_round)
    result_number.append(rounds)
```

## 13 Why This Matters

This example connects several important ideas:

- probability,
- loops,
- lists of lists,
- simulations,
- plotting,
- random walks.

It also gives a real reason to store data during a simulation.

## 14 Key Takeaways

- The dice game has equal probabilities of winning or losing one credit.
- Repeated re-rolls naturally lead to geometric series.
- The game behaves like a random walk.
- Simulations can be used to study behavior that is difficult to understand from a single run.
- Lists are useful for storing simulation histories.